

Volumes by the Disc and Washer Methods

ANSWER KEY



The disc method (about the x-axis)

- 1 Find the volume of the solid formed by revolving $y = \sqrt{x}$ about the x-axis on $[0, 4]$.

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = \pi(8) = 8\pi.$$

- 2 Find the volume of the solid formed by revolving $y = x^2$ about the x-axis on $[0, 2]$.

$$V = \pi \int_0^2 (x^2)^2 dx = \pi \int_0^2 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^2 = \pi \cdot \frac{32}{5} = \frac{32\pi}{5}.$$

The disc method (about the y-axis)

- 3 Find the volume formed by revolving $x = y^2$ about the y-axis on $y \in [0, 2]$.

$$V = \pi \int_0^2 (y^2)^2 dy = \pi \int_0^2 y^4 dy = \pi \left[\frac{y^5}{5} \right]_0^2 = \frac{32\pi}{5}.$$

The washer method

- 4 Find the volume formed by revolving the region between $y = x$ and $y = x^2$ (on $[0, 1]$) about the x-axis.

Outer radius $R = x$ (larger on $[0, 1]$), inner radius $r = x^2$.

$$V = \pi \int_0^1 [x^2 - (x^2)^2] dx = \pi \int_0^1 (x^2 - x^4) dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}.$$

- 5 Find the volume formed by revolving the region between $y = \sqrt{x}$ and $y = x$ about the x-axis on $[0, 1]$.

On $[0, 1]$, $\sqrt{x} \geq x$, so outer radius $R = \sqrt{x}$, inner radius $r = x$.

$$V = \pi \int_0^1 [x - x^2] dx = \pi \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \pi \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{\pi}{6}.$$

Volumes about a horizontal or vertical line (not an axis)

- 6 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 4$ about the line $y = 4$.

Intersections at $x = \pm 2$. Radius = $4 - x^2$ (distance from curve to the line $y = 4$).

$$V = \pi \int_{-2}^2 (4 - x^2)^2 dx = \pi \int_{-2}^2 (16 - 8x^2 + x^4) dx = \pi \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_{-2}^2 = 2\pi \left(32 - \frac{64}{3} + \frac{32}{5} \right) = 2\pi \cdot \frac{480 - 320 + 96}{15} = 2\pi \cdot \frac{256}{15} = \frac{512\pi}{15}.$$

Volumes with known cross-sections

- 7 The base of a solid is the region between $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x-axis are squares. Find the volume.

Side length of each square = $4 - x^2$.

$$V = \int_{-2}^2 (4 - x^2)^2 dx = \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_{-2}^2 = 2 \left(32 - \frac{64}{3} + \frac{32}{5} \right) = 2 \cdot \frac{256}{15} = \frac{512}{15} \text{ (same integrand as the washer-about-} y = 4 \text{ problem, but without the extra } \pi \text{ since cross-sections are squares, not circles).}$$

- 8 The base of a solid is a circle of radius 2 ($x^2 + y^2 = 4$). Cross-sections perpendicular to the x -axis are semicircles. Find the volume.

At each x , the base chord has length $2y = 2\sqrt{4 - x^2}$, which is the diameter of the semicircular cross-section, so its radius is $\sqrt{4 - x^2}$. Semicircle area $= \frac{1}{2}\pi(\sqrt{4 - x^2})^2 = \frac{\pi}{2}(4 - x^2)$.

$$V = \int_{-2}^2 \frac{\pi}{2}(4 - x^2) dx = \frac{\pi}{2} \left[4x - \frac{x^3}{3} \right]_{-2}^2 = \frac{\pi}{2} \cdot 2 \left(8 - \frac{8}{3} \right) = \pi \cdot \frac{16}{3} = \frac{16\pi}{3}.$$

- 9 Set up (but do not evaluate) the integral for the volume of the solid whose base is the region bounded by $y = \sin x$ and the x -axis on $[0, \pi]$, with cross-sections perpendicular to the x -axis that are equilateral triangles.

Each equilateral triangle has side length $s = \sin x$, so its area is $\frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}\sin^2 x$. $V = \int_0^{\pi} \frac{\sqrt{3}}{4}\sin^2 x dx$.

Volume by revolving about the y -axis (washer method)

- 10 Find the volume formed by revolving the region between $x = y^2$ and $x = 2y$ about the y -axis.

Intersections: $y^2 = 2y \Rightarrow y(y - 2) = 0$: $y = 0, 2$. On $[0, 2]$, outer radius $R = 2y$, inner radius $r = y^2$.

$$V = \pi \int_0^2 [(2y)^2 - (y^2)^2] dy = \pi \int_0^2 (4y^2 - y^4) dy = \pi \left[\frac{4y^3}{3} - \frac{y^5}{5} \right]_0^2 = \pi \left(\frac{32}{3} - \frac{32}{5} \right) = \pi \cdot \frac{160 - 96}{15} = \frac{64\pi}{15}.$$

Visualizing a solid of revolution's generating region

- 11 The graph shows $y = \sqrt{x}$ on $[0, 4]$ from the first disc-method problem. Sketching this region mentally revolved about the x -axis produces a solid — use the volume 8π found earlier to find its average cross-sectional area over the interval, and compare that to the cross-sectional area at the midpoint $x = 2$.
Average cross-sectional area $= \frac{V}{4} = \frac{8\pi}{4} = 2\pi$ (dividing volume by the interval's length). At $x = 2$, the cross-section is a disc of radius $\sqrt{2}$, area $= \pi(\sqrt{2})^2 = 2\pi$ — matching the average exactly, since the cross-sectional area πx is linear in x , so its average equals its midpoint value.

A capstone washer-method problem

- 12 Find the volume formed by revolving the region between $y = 4 - x^2$ and $y = x^2$ about the x -axis (using the region where $4 - x^2 \geq x^2$).

Intersections: $4 - x^2 = x^2 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$. Outer radius $R = 4 - x^2$, inner radius $r = x^2$.

$$V = \pi \int_{-\sqrt{2}}^{\sqrt{2}} [(4 - x^2)^2 - (x^2)^2] dx = \pi \int_{-\sqrt{2}}^{\sqrt{2}} (16 - 8x^2) dx = \pi \left[16x - \frac{8x^3}{3} \right]_{-\sqrt{2}}^{\sqrt{2}} = 2\pi \left(16\sqrt{2} - \frac{16\sqrt{2}}{3} \right) = 2\pi \cdot \frac{32\sqrt{2}}{3} = \frac{64\sqrt{2}\pi}{3}.$$

Volumes with rectangular cross-sections

- 13** The base of a solid is the region bounded by $y = x^2$ and $y = 1$. Cross-sections perpendicular to the x -axis are rectangles with height twice the base. Find the volume.

At each $x \in [-1, 1]$, the base width is $1 - x^2$, so the rectangle's area is $2(1 - x^2)^2$.

$$V = \int_{-1}^1 2(1 - x^2)^2 dx = 2 \int_{-1}^1 (1 - 2x^2 + x^4) dx = 2 \left[x - \frac{2x^3}{3} + \frac{x^5}{5} \right]_{-1}^1 = 2 \cdot 2 \left(1 - \frac{2}{3} + \frac{1}{5} \right) = 4 \cdot \frac{15 - 10 + 3}{15} = 4 \cdot \frac{8}{15} = \frac{32}{15}$$

A capstone volume application

- 14** A bowl is formed by revolving $y = \frac{x^2}{4}$ (for $0 \leq x \leq 4$, where x is horizontal distance and y is depth from the rim) about the y -axis. Set up (do not evaluate) the disc-method integral for the volume of the bowl in terms of y , using the fact that $x^2 = 4y$.

$$V = \pi \int_0^4 x^2 dy = \pi \int_0^4 4y dy \text{ (using } y \text{ as the integration variable and } x^2 = 4y \text{ from solving the original equation for } x^2 \text{).}$$

- 15** Find the volume formed by revolving $y = e^x$ about the x -axis on $[0, 1]$.

$$V = \pi \int_0^1 (e^x)^2 dx = \pi \int_0^1 e^{2x} dx = \pi \left[\frac{1}{2} e^{2x} \right]_0^1 = \frac{\pi}{2} (e^2 - 1).$$

- 16** A vase is designed by revolving $y = \frac{1}{2}x + 1$ about the x -axis on $[0, 6]$, where x is the height (inches) and y is the radius. Find the volume of the vase.

$$V = \pi \int_0^6 \left(\frac{x}{2} + 1 \right)^2 dx = \pi \int_0^6 \left(\frac{x^2}{4} + x + 1 \right) dx = \pi \left[\frac{x^3}{12} + \frac{x^2}{2} + x \right]_0^6 = \pi(18 + 18 + 6) = 42\pi \text{ in}^3.$$

- 17** For the region between $y = x$ and $y = x^3$ on $[0, 1]$ (where $y = x \geq y = x^3$), complete each step for the solid formed by revolving this region about the x -axis:

a Identify the outer radius R and inner radius r . Outer radius $R = x$ (the larger function on $[0, 1]$), inner radius $r = x^3$.

b Set up the washer-method integral. $V = \pi \int_0^1 [x^2 - x^6] dx$.

c Evaluate the integral. $V = \pi \left[\frac{x^3}{3} - \frac{x^7}{7} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{7} \right) = \frac{4\pi}{21}$.